1. OPTIONS, INTEREST, NO ARBITRAGE

European payoff at expiry T

Call: $C_T = (S_T - K)^+ = \max(S_T - K, 0)$. Put: $P_T = (K - S_T)^+$. European: exercise only at T. American: exercise any $t \in [0, T]$.

Continuously compounded risk-free account

$B(T) = B(0)e^{rT}$, so $PV(FV) = FV\, e^{-rT}$. Over one lattice step: growth $a = e^{r\Delta t}$, discount $e^{-r\Delta t}$.

No-arbitrage principle

A risk-free portfolio must earn the risk-free rate. Arbitrage: zero net initial wealth, no chance of loss, positive chance/certain gain. Option price is determined by replication/hedging, *not* physical up/down probabilities.

2. TWO-STATE TREE: MASTER RECIPE

At $t = 0$: stock S_0. At T: (S_u, V_u) or (S_d, V_d).

Hedging portfolio (lecture convention)

Long δ shares, short one option: $\Pi = \delta S - V$. Choose

$$\boxed{\delta = \frac{V_u - V_d}{S_u - S_d}}$$

so $\Pi_u = \delta S_u - V_u = \Pi_d = \delta S_d - V_d =: \Pi_T$. Then

$$\Pi_0 = e^{-rT}\Pi_T, \qquad \boxed{V_0 = \delta S_0 - e^{-rT}\Pi_T}.$$

If using $P = V - \delta S$, all signs flip consistently, but the same δ results.

Equivalent risk-neutral formula

For multiplicative moves $S_u = uS_0$, $S_d = dS_0$:

$$\boxed{p^* = \frac{e^{rT} - d}{u - d}}, \quad q^* = 1 - p^*, \quad \boxed{V_0 = e^{-rT}(p^* V_u + q^* V_d)}.$$

General form: $p^* = (e^{rT}S_0 - S_d)/(S_u - S_d)$. No-arbitrage requires $d < e^{rT} < u$ (so $0 < p^* < 1$).

Mispricing arbitrage shortcut

If market $V^* > V_{fair}$: sell overpriced option + buy replicating hedge, lock $V^* - V_{fair}$. If $V^* < V_{fair}$: buy option + short replicating hedge, lock $V_{fair} - V^*$.

3. PROBABILITY / NORMAL FACTS

$Y \sim N(\mu, \beta^2)$:

$$f_Y(y) = \frac{1}{\sqrt{2\pi\beta^2}} e^{-(y-\mu)^2/(2\beta^2)}, \quad \mathbb{E}Y = \mu, \quad \operatorname{Var} Y = \beta^2.$$

$\operatorname{Var} X = \mathbb{E}[X^2] - (\mathbb{E}X)^2$. Independent X, Y: $\mathbb{E}[XY] = \mathbb{E}X\mathbb{E}Y$, $\operatorname{Var}(X + Y) = \operatorname{Var} X + \operatorname{Var} Y$.

Standardize: if $Y \sim N(\mu, s^2)$, then $\mathbb{P}(Y \le y) = \Phi((y - \mu)/s)$. For $Z_t \sim N(0, t)$: $\mathbb{E}[Z_t^2] = t$, $\mathbb{E}[Z_t^4] = 3t^2$.

4. BROWNIAN MOTION / SDE BASICS

$$\boxed{dX = \alpha\, dt + \sigma\, dZ}, \qquad dZ = \phi\sqrt{dt}, \quad \phi \sim N(0, 1).$$

$$\mathbb{E}[dZ] = 0, \quad \operatorname{Var}(dZ) = dt, \quad \mathbb{E}[dX] = \alpha dt, \quad \operatorname{Var}(dX) = \sigma^2 dt.$$

Over finite t (constant α, σ):

$$X(t) = X_0 + \alpha t + \sigma(Z_t - Z_0), \quad Z_t - Z_0 \sim N(0, t),$$

thus $\boxed{X(t) \sim N(X_0 + \alpha t, \sigma^2 t)}$. Brownian paths are continuous but non-differentiable; $dZ/dt \sim 1/\sqrt{dt} \to \infty$.

5. DISCRETE RANDOM WALK → SDE

One step: $\Delta X = +\Delta h$ w.p. p, $-\Delta h$ w.p. $q = 1 - p$.

$$\mathbb{E}[\Delta X] = (p - q)\Delta h, \quad \mathbb{E}[(\Delta X)^2] = (\Delta h)^2,$$

$$\operatorname{Var}(\Delta X) = 4pq(\Delta h)^2.$$

After $n = t/\Delta t$ independent steps:

$$\mathbb{E}[X_n - X_0] = \frac{t}{\Delta t}(p - q)\Delta h, \quad \operatorname{Var}(X_n - X_0) = \frac{t}{\Delta t} 4pq(\Delta h)^2.$$

Probability of j up moves:

$$\mathbb{P}(j) = \binom{n}{j} p^j q^{n-j}, \quad X_n = X_0 + (2j - n)\Delta h.$$

To converge to $dX = \alpha dt + \sigma dZ$ choose

$$\boxed{\Delta h = \sigma\sqrt{\Delta t}}, \qquad \boxed{p = \tfrac{1}{2}\left(1 + \frac{\alpha}{\sigma}\sqrt{\Delta t}\right)}, \quad q = 1 - p.$$

Then $\mathbb{E}[X_n - X_0] = \alpha t$ and $\operatorname{Var}(X_n - X_0) = \sigma^2 t\left(1 - \frac{\alpha^2}{\sigma^2}\Delta t\right) \to \sigma^2 t$.

6. ITÔ CALCULUS: RULES TO MEMORIZE

Quadratic variation / order rules

$$\boxed{(dZ)^2 = dt}, \qquad dt\, dZ = 0, \qquad (dt)^2 = 0$$

when retaining terms through order dt. If $dX = a\, dt + b\, dZ$, then

$$\boxed{(dX)^2 = b^2 dt}.$$

Cross variation: if $dX = a_X dt + b_X dZ_X$, $dY = a_Y dt + b_Y dZ_Y$, then

$$\boxed{dX\, dY = \rho\, b_X b_Y dt}, \qquad dZ_X dZ_Y = \rho\, dt.$$

Same Wiener: $\rho = 1$; independent Wiener processes: $\rho = 0$. Also $\sum_i (\Delta Z_i)^2 \to T$ on $[0, T]$.

Itô stochastic integral

$$\int_0^t c(X_s, s) dZ_s = \lim_{\Delta t \to 0} \sum_j c(X_{t_j}, t_j)\Delta Z_j$$

uses the *left endpoint* (no peeking into the future).

Itô's Lemma

If $dX = a(X, t)dt + b(X, t)dZ$ and $F = F(X, t)$,

$$\boxed{dF = \left(F_t + aF_X + \tfrac{1}{2}b^2 F_{XX}\right) dt + bF_X dZ}.$$

Recipe: compute F_X, F_{XX}, F_t; identify a, b; substitute; group dt (drift) and dZ (random) terms.

Product / integration by parts

$$\boxed{d(XY) = X\, dY + Y\, dX + dX\, dY}.$$

Hence $\int X\, dY = [XY]_0^T - \int Y\, dX - \int dX\, dY$. For standard Brownian $Z(0) = 0$:

$$\boxed{\int_0^T Z\, dZ = \tfrac{1}{2}Z(T)^2 - \tfrac{1}{2}T}.$$

If $dX = \mu dt + \sigma(X, t)dZ$, then $\boxed{\int_0^T (dX)^2 = \int_0^T \sigma(X, t)^2 dt}$.

7. GBM AND HIGH-VALUE ITÔ TRANSFORMS

Stock GBM:

$$\boxed{dS = \mu S dt + \sigma S dZ} \quad \Leftrightarrow \quad dS/S = \mu dt + \sigma dZ.$$

For $X = \ln S$:

$$\boxed{d\ln S = (\mu - \tfrac{1}{2}\sigma^2)dt + \sigma dZ}.$$

So (constant μ, σ)

$$\boxed{S_t = S_0 \exp\left[(\mu - \tfrac{1}{2}\sigma^2)t + \sigma(Z_t - Z_0)\right]}.$$

Thus $\ln(S_t/S_0) \sim N((\mu - \sigma^2/2)t, \sigma^2 t)$,

$$\boxed{\mathbb{E}[S_t] = S_0 e^{\mu t}}, \qquad \boxed{\operatorname{Var}(S_t) = S_0^2 e^{2\mu t}(e^{\sigma^2 t} - 1)}.$$

Power shortcut for $Y = S^k$:

$$\boxed{dY = \left[k\mu + \tfrac{1}{2}k(k-1)\sigma^2\right] Y dt + k\sigma Y dZ}.$$

Example $k = 1/2$: drift $= (\mu/2 - \sigma^2/8)\sqrt{S}$, random coeff. $= (\sigma/2)\sqrt{S}$. For constant μ, σ:

$$\boxed{\mathbb{E}[S_t^k] = S_0^k \exp\left(k\mu t + \tfrac{1}{2}k(k-1)\sigma^2 t\right)}.$$

Quadratic variation / A1 identity

For independent $\Delta Z_i = \phi_i\sqrt{\Delta t}$, $\phi_i \sim N(0, 1)$, and $I(T, \Delta t) = \sum_{i=0}^{N-1} (\Delta Z_i)^2$, $\Delta t = T/N$:

$$\mathbb{E}[(\Delta Z_i)^2] = \Delta t, \qquad \operatorname{Var}((\Delta Z_i)^2) = 2(\Delta t)^2,$$

$$\boxed{\mathbb{E}[I] = T}, \qquad \boxed{\operatorname{Var}(I) = 2T\Delta t \to 0}.$$

So $\sum (\Delta Z_i)^2 \to T$, the finite-time version of $(dZ)^2 = dt$.

8. RISK-NEUTRAL WORLD

Under GBM, no-arbitrage pricing can be done in the Q (risk-neutral) world:

$$\boxed{dS = rSdt + \sigma S dZ} \qquad \text{(replace physical drift } \mu \text{ by } r\text{)}.$$

For $X = \ln S$ under Q:

$$dX = (r - \tfrac{1}{2}\sigma^2)dt + \sigma dZ.$$

With $\tau = T - t$:

$$\boxed{\ln S_T \mid S_t = S \sim N\left(\ln S + (r - \tfrac{1}{2}\sigma^2)\tau,\ \sigma^2\tau\right)},$$

so $S_T = S\exp[(r - \sigma^2/2)\tau + \sigma\sqrt{\tau}\,\phi]$, $\phi \sim N(0,1)$.
European payoff $H(S_T)$:

$$\boxed{V(S,t) = e^{-r(T-t)}\mathbb{E}^Q[H(S_T) \mid S_t = S]}.$$

Risk-neutral probabilities are pricing/hedging coefficients, not forecasts of real stock moves.

9. NO-ARBITRAGE BINOMIAL LATTICE

N steps, $\Delta t = T/N$:

$$\boxed{u = e^{\sigma\sqrt{\Delta t}}, \quad d = e^{-\sigma\sqrt{\Delta t}} = 1/u, \quad a = e^{r\Delta t}}$$

$$\boxed{p^* = \frac{a-d}{u-d}}, \qquad q^* = 1 - p^*.$$

No-arbitrage/stability of this tree requires $d < a < u$ (equivalently $0 < p^* < 1$).
Stock moves: $S_{j+1}^{n+1} = uS_j^n$, $S_j^{n+1} = dS_j^n$ and directly

$$\boxed{S_j^n = S_0 u^j d^{n-j} = S_0 e^{(2j-n)\sigma\sqrt{\Delta t}}}.$$

Node hedge (delta):

$$\boxed{\delta_j^n = \frac{V_{j+1}^{n+1} - V_j^{n+1}}{S_{j+1}^{n+1} - S_j^{n+1}} \to V_S}.$$

European rollback
Set terminal payoff $V_j^N = (S_j^N - K)^+$ (call) or $(K - S_j^N)^+$ (put), then for $n = N-1, \ldots, 0$:

$$\boxed{V_j^n = e^{-r\Delta t}\left(p^* V_{j+1}^{n+1} + q^* V_j^{n+1}\right)}.$$

American rollback
Continuation: $V_c = e^{-r\Delta t}(p^* V_{j+1}^{n+1} + q^* V_j^{n+1})$. Exercise payoff $H(S_j^n)$. Then

$$\boxed{V_j^n = \max\{V_c, H(S_j^n)\}}.$$

At every node: compare **exercise now** vs **continue**.
Alternative drifting lattice (stable)
Risk-neutral log drift $\alpha = r - \sigma^2/2$, choose $p = 1/2$ and

$$u = \exp[\sigma\sqrt{\Delta t} + (r - \sigma^2/2)\Delta t], \quad d = \exp[-\sigma\sqrt{\Delta t} + (r - \sigma^2/2)\Delta t].$$

10. PUT-CALL PARITY

European, same S, K, T:

$$\boxed{C(S,t) - P(S,t) + Ke^{-r(T-t)} = S}.$$

On lattice node (j,n), $T = N\Delta t$:

$$\boxed{C_j^n - P_j^n + Ke^{-(N-n)r\Delta t} = S_j^n}.$$

At maturity this reduces to $(S-K)^+ - (K-S)^+ + K = S$.

11. DYNAMIC PROGRAMMING / OPTIMAL STOPPING

Principle of optimality: from any point on an optimal path, the remaining decisions are optimal for the resulting subproblem. Work *backwards*.
Generic stopping problem:

$$\boxed{V_k = \mathbb{E}[\max\{W_k, V_{k+1}\}]}$$

(no extra costs), so stop iff $W_k > V_{k+1}$. Equivalent lecture notation:

$$p_k = \mathbb{P}(W_k > V_{k+1}), \quad \bar{W}_k = \mathbb{E}[W_k \mid W_k > V_{k+1}],$$

$$\boxed{V_k = p_k\bar{W}_k + (1-p_k)V_{k+1}}.$$

If taking a new action/roll costs c, subtract that cost exactly when the action is taken.
Uniform $U[a,b]$ shortcut
$\mathbb{E}W = (a+b)/2$. For $a \le c \le b$:

$$\mathbb{P}(W > c) = \frac{b-c}{b-a}, \qquad \mathbb{E}[W \mid W > c] = \frac{b+c}{2}.$$

This is the fastest way to do the marriage/practice stopping questions.

12. CONVERGENCE / ORDER OF ACCURACY

A method is pth-order accurate if

$$\boxed{V_{\Delta t} = V_{exact} + C(\Delta t)^p + O((\Delta t)^{p+1})}.$$

Halving Δt reduces leading error by 2^p. Without knowing V_{exact}:

$$\boxed{R = \frac{V_{\Delta t/2} - V_{\Delta t}}{V_{\Delta t/4} - V_{\Delta t/2}} \to 2^p}, \qquad p \approx \log_2|R|.$$

No-arbitrage lattice is generally first order: $p = 1$, so $R \to 2$. Payoff kinks can make convergence erratic unless a terminal node lands at $S = K$ or payoff is smoothed.
Richardson extrapolation (first-order)
If $V_h = V + Ch + O(h^2)$:

$$\boxed{V_{exact} \approx 2V_{h/2} - V_h = V_{h/2} + (V_{h/2} - V_h)}.$$

13. PAYOFF SMOOTHING

Let $X = \ln S$, terminal nodes spaced by $2h$ where $h = \sigma\sqrt{\Delta t}$. Replace payoff $g(X)$ by its cell average:

$$\boxed{\hat{g}(X_j) = \frac{1}{2h}\int_{X_j-h}^{X_j+h} g(x)\,dx}.$$

For regular options: $g_C(x) = (e^x - K)^+$, $g_P(x) = (K - e^x)^+$.
Digital put (A2 high-value formula)
Payoff $g = 1$ if $S < K$, else 0. Let $x_j = \ln S_j$, $k = \ln K$:

$$\boxed{\hat{g}(x_j) = \begin{cases} 1, & x_j + h < k, \\ 0, & x_j - h \ge k, \\ \dfrac{k - x_j + h}{2h}, & x_j - h < k \le x_j + h. \end{cases}}$$

Equivalent crossing expression: $[\ln(K/S_j) + h]/(2h)$.

14. BLACK-SCHOLES PDE: DERIVATION SKELETON

Assumptions used in lecture: GBM stock, known risk-free r, can borrow/lend at r, no transaction costs, known volatility, short selling allowed.
Take $V = V(S,t)$ and portfolio

$$\boxed{\Pi = V - \delta S}, \qquad d\Pi = dV - \delta\, dS$$

with δ held fixed over the tiny interval. GBM: $dS = \mu S dt + \sigma S dZ$. Itô on V:

$$dV = \left(V_t + \mu S V_S + \tfrac{1}{2}\sigma^2 S^2 V_{SS}\right)dt + \sigma S V_S dZ.$$

Therefore

$$d\Pi = \left[V_t + \mu S(V_S - \delta) + \tfrac{1}{2}\sigma^2 S^2 V_{SS}\right]dt + \sigma S(V_S - \delta)dZ.$$

Choose the hedge

$$\boxed{\delta = V_S}$$

to kill randomness. Then $d\Pi = (V_t + \frac{1}{2}\sigma^2 S^2 V_{SS})dt$. Since risk-free, $d\Pi = r\Pi dt$ and $\Pi = V - SV_S$. Hence

$$\boxed{V_t + \tfrac{1}{2}\sigma^2 S^2 V_{SS} + rSV_S - rV = 0}.$$

Key fact: μ cancels. Solve backward from terminal payoff at T. With time-to-expiry $\tau = T - t$:

$$\boxed{V_\tau = \tfrac{1}{2}\sigma^2 S^2 V_{SS} + rSV_S - rV}, \qquad V(S,0) = H(S).$$

American options additionally satisfy $\boxed{V(S,\tau) \ge H(S)}$; exercise where equality holds, continue where the PDE holds.
Continuous-time delta hedge
If short one option: hold V_S shares and bank position $V - SV_S$ (negative means borrow). Rebalance continuously so shares always equal updated V_S. Self-financing idealized hedge.

15. EXAM SPEED RECIPES

One-step option: 1) terminal stock/payoffs, 2) p^* or δ, 3) discount, 4) if mispriced, trade opposite to mispricing.
Brownian mean/variance: deterministic shifts change mean only; multiplying by c multiplies variance by c^2; $Z_t \sim N(0,t)$.
Itô transform: write a, b first, compute 3 derivatives, plug lemma, use $(dZ)^2 = dt$.
Lattice: terminal payoff first, then rollback; American adds max(continuation, exercise).
DP: solve final stage exactly, use its value as continuation threshold one stage earlier, repeat backward.
Black-Scholes derivation: Itô $dV \to$ form $V - \delta S \to$ set $\delta = V_S \to$ risk-free growth $r\Pi \to$ PDE.

16. COMMON TRAPS

- Do not use real-world up/down probabilities to price a hedged option.
- Convert months to years when r is annual; continuous discount is e^{-rT}.
- σ is standard deviation rate; variance scales as $\sigma^2 t$.
- In Itô, never drop the $\frac{1}{2}b^2 F_{XX}$ term.
- dZ is order $\sqrt{dt}$, so $(dZ)^2$ is order dt and survives.
- American decision is local at every node, not only at expiry.
- Ratio test uses successive *differences*; for first order expect ratio 2 (sign may oscillate before asymptotic regime).

17. LAST-MINUTE IDENTITIES

$\mathbb{E}[aX + b] = a\mathbb{E}[X] + b$, $\mathrm{Var}(aX + b) = a^2\,\mathrm{Var}(X)$. For independent increments, variances add.
If $Y = Z_t - ct$, then $\mathbb{E}[Y] = -ct$, $\mathrm{Var}(Y) = t$.
Risk-free growth/discount: $FV = PV\,e^{r\tau}$, $PV = FV\,e^{-r\tau}$.
At expiry: call slope is 0 below K, 1 above K; put slope is -1 below K, 0 above K. The kink at K is why raw lattice convergence can oscillate.